AMI – Amazon Machine Image

❖ What is an AMI?

➢ An Amazon Machine Image (AMI) is a template that contains the software configuration required to launch an EC2 instance.

▪ **It includes:**

- **Operating system** (e.g., Amazon Linux, Ubuntu, Windows)
- **Pre-installed software** (web servers, databases, custom applications)
- **Settings and permissions**
- **Root volume and data volumes**

When you launch an instance, you select an AMI to define the initial state of the machine

❖ Types of AMIs

➢ Amazon provides multiple types of AMIs depending on the use case:

Type	Description
AWS-provided AMIs	Official images provided and maintained by AWS (e.g.Amazon Linux, Windows Server)
Marketplace AMIs	Third-party AMIs available for purchase or subscription in AWS Marketplace
Community AMIs	AMIs shared publicly by AWS users (may not beverified or supported)
Custom AMIs	User-created AMIs built from an existing EC2 instance

❖ What is a Launch Template?

➢ A Launch Template in Amazon EC2 is a reusable configuration for launching EC2 instances.

▪ It **defines** the instance settings such as

- ***AMI ID***
- ***instance type***
- ***key pair***
- ***security groups***
- ***storage configuration***
- ***tags, and network interfaces***

❖ Why Use Launch Templates?

➢ Launch Templates in AWS simplify and standardize the process of launching EC2 instances by saving all the configuration details in one reusable template.

- Key Benefits:
    - Reusability
    - Version Control
    - Integration with Other AWS Services
    - Parameter Consistency
    - Overrides Supported
    - Automation Friendly

❖ How do you migrate an EC2 instance from one region to another?

1. Create an Amazon Machine Image (AMI) of the existing instance.
2. Copy the AMI to the target region.
3. Launch a new EC2 instance from the copied AMI in the target region

❖ EC2 Instance Purchasing Options: *(Day 5 on LMS see)*

Option	**Description**	**Best For**
On-Demand	Pay per hour/second, no commitment	Short-term, unpredictable workloads
Reserved Instances (RI)	Commit for 1 or 3 years, and get up to 75% discount	Steady or long-term workloads
Spot Instances	Buy unused EC2 at up to 90% discount, can be interrupted anytime	Flexible, fault-tolerant tasks
Savings Plans	Commit to spend ($/hr) over 1 or 3 years, flexible across instance type	Cost savings with flexibility
Capacity Reservations	Reserve capacity in a specific AZ without commitment	Guaranteed capacity in AZ

Interview One-liners:

- ✅ **On-Demand:** *"Best for short-term or testing workloads with no commitment."*
- ✅ **Reserved:** *"Good for predictable, long-term use with big cost savings."*
- ✅ **Spot:** *"Cheap option for flexible workloads like batch jobs or data analysis."*
- ✅ **Savings Plan:** *"Offers flexibility and savings like Reserved Instances, but more flexible."*

[General Interview Questions]

Q1. What are EC2 Capacity Reservations ?

➢ EC2 Capacity Reservation allows you to reserve compute capacity (specific instance type in a specific Availability Zone) without long-term commitment.

- Ensures **capacity is available** when you need it.
- You pay **On-Demand price** for reserved capacity.
- Useful for **critical workloads** or disaster recovery setups.

Q2. How do they differ from Capacity Reservation & Reserved Instances

Feature	**Capacity Reservations**	**Reserved Instances (RIs)**
Purpose	Reserve capacity in a specific AZ	Save cost with long-term

		commitment
Pricing	Pay On-Demand rate	Pay upfront or partially, up to 75% discount
Commitment	✗ No time commitment	☑ 1 or 3-year commitment
Capacity Guarantee	☑ Yes (guarantees instance availability)	✗ No (only cost-saving, doesn't guarantee capacity)
Flexibility	Flexibility	More flexible, but doesn't hold capacity

Q3. How EC2 Achieves High Network Throughput and what features help in optimizing network performance.

1. **Elastic Network Adapter (ENA)**
   - ENA supports high-performance networking with up to **100 Gbps** throughput.
   - Used for modern instance types (e.g., C5n, M5n, R5n).
2. **EFA (Elastic Fabric Adapter)**
   - Specialized for **high-performance computing (HPC)** and low-latency communication.
3. **Enhanced Networking**
   - Provides higher IOPS, lower latency, and more packets per second (PPS).
4. **Placement Groups**
   - **Cluster Placement Group:** Places instances physically close together to reduce latency and increase throughput.

Q4. What is a Spot Fleet, and how does it help in optimizing costs?

- A **Spot** Fleet is a collection of EC2 Spot Instances and, optionally, On-Demand Instances that AWS launched based on a specified target capacity.
- **It helps** in achieving cost optimization by automatically selecting the lowest-priced Spot Instances from different Availability Zones.

Q5. How do you migrate an EC2 instance from one region to another?

1. **Create an Amazon Machine Image** (AMI) of the existing instance.
2. **Copy the AMI** to the target region.
3. Launch a new EC2 instance from the **copied AMI** in the **target region.**

Q6. How do you prevent data loss in EC2 instances?

- To prevent **data loss** in EC2, use **EBS** for persistent storage, take regular snapshots, store important data in **S3**,
- use AMIs for backups, enable **termination protection**, and monitor your instance health with **CloudWatch**.
- These practices ensure high durability, availability, and quick recovery.

Q7. What are the major differences between AWS EC2 Auto Scaling and AWS Elastic Load Balancer (ELB)?

- **Auto Scaling:** Dynamically adjusts the number of EC2 instances based on traffic.
- **Elastic Load Balancer:** Distributes incoming traffic among multiple EC2 instances.

Q8. Explain the concept of EC2 Dedicated Hosts and their use cases.

- An EC2 ***Dedicated Host*** is a physical server fully dedicated to your use.
- Used for regulatory compliance**, Bring Your Own License** (BYOL) scenarios, and predictable workloads.

Q9. What is the difference between EC2 Hibernate and EC2 Stop?

- EC2 **Stop** shuts down the instance and loses memory state, while

- **Hibernate** saves the RAM and resumes the instance from where it left off. Hibernate is faster to restart **but** uses more EBS storage.